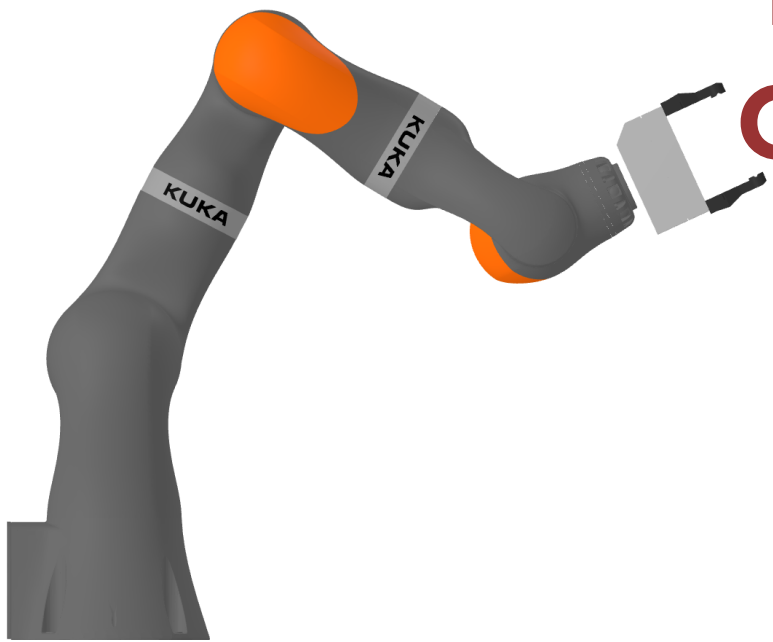
Manipulation in Clutter

Part 2:

Grasp Selection



MIT 6.4210/2

Robotic Manipulation

Fall 2025, Lecture 8

Massachusetts Institute of Technology

Course Roadmap

1. Basic pick and place:
    - single *known* object
    - assuming *known* pose
2. Geometric perception
    - single *known* object
    - *unknown* pose
3. **This week**
    - many/diverse *unknown* objects
    - *unknown* poses
    - simple task: "clutter clearing"

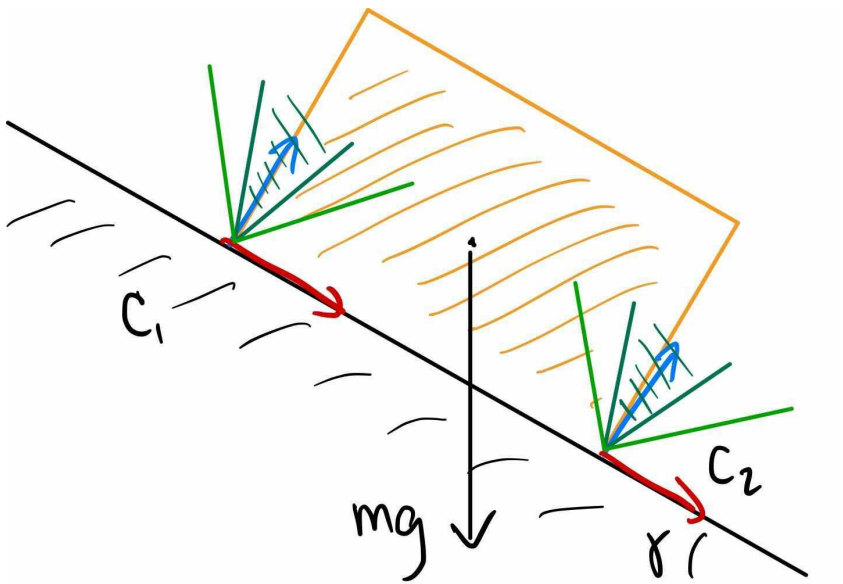
"Clutter clearing" (an early project at TRI)

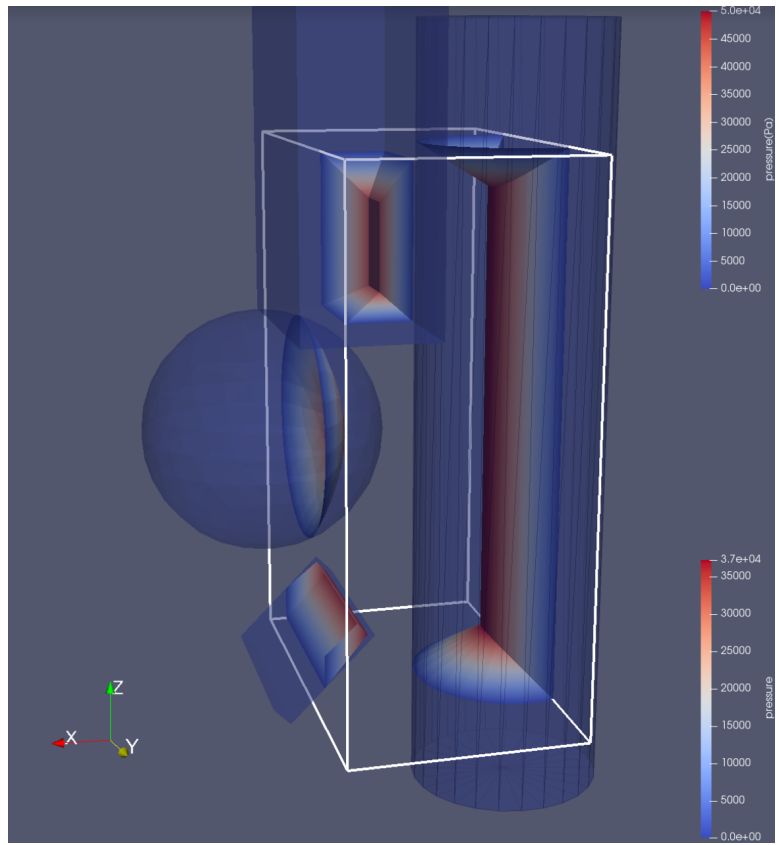


Initial conditions using simulation

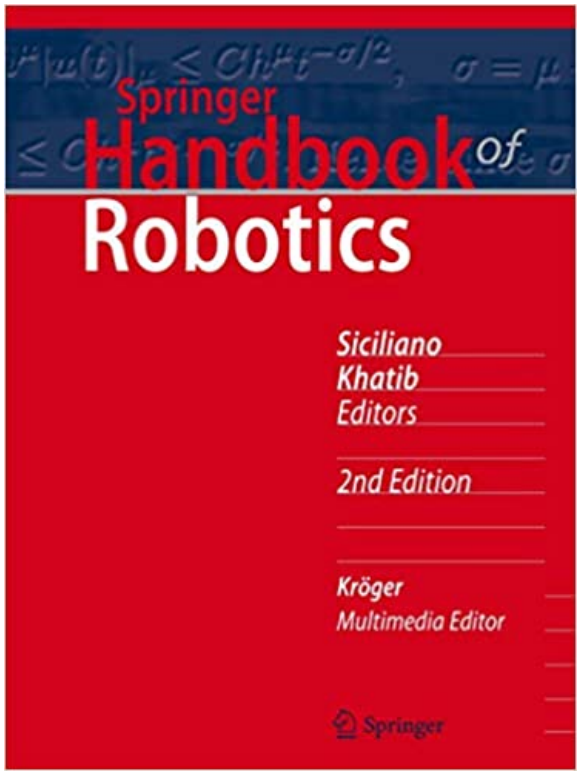
https://manipulation.csail.mit.edu/data/ falling_bricks_2d.html?height=400px

Contact mechanics (friction cones, hydroelastic)





Grasp analysis / optimization



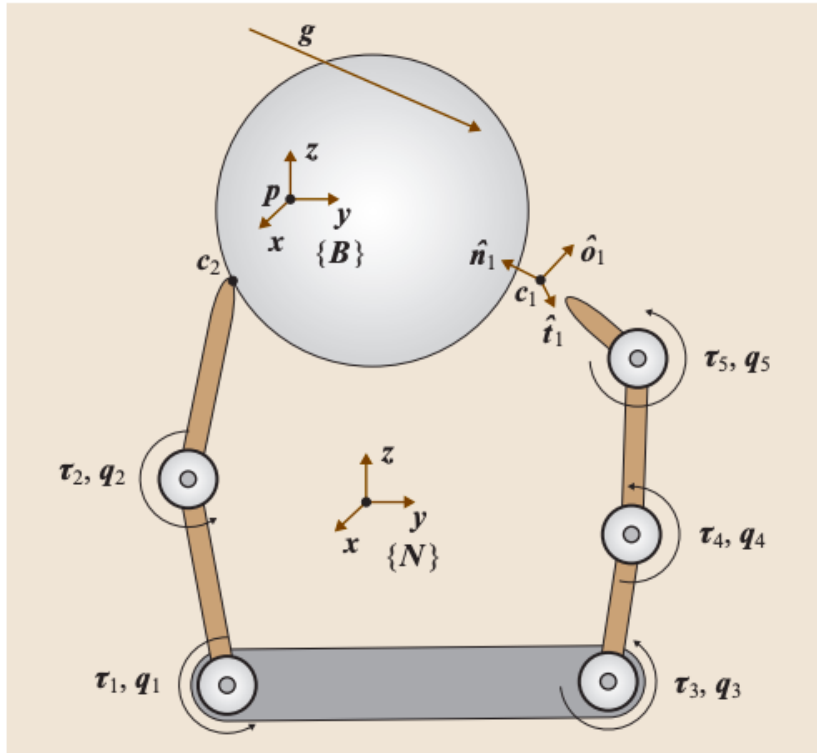
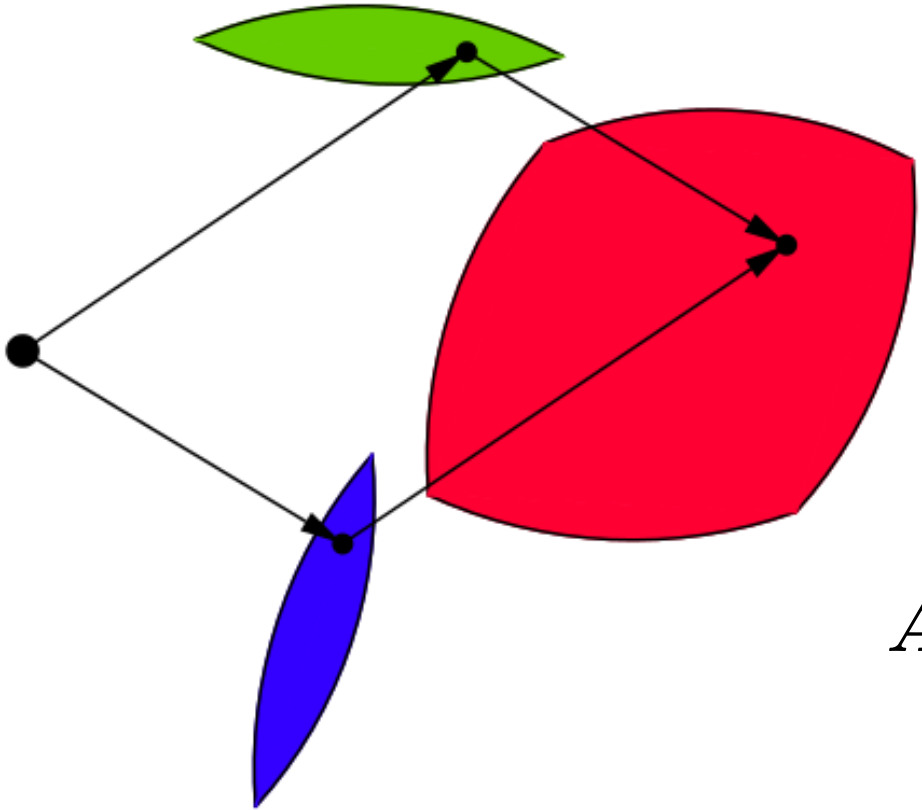


Fig. 28.2 Main quantities for grasp analysis

Grasp closures

- Form closure
  - There is no direction that the object can move if the fingers are fixed.
- Force closure
  - The grasp can resist all wrenches

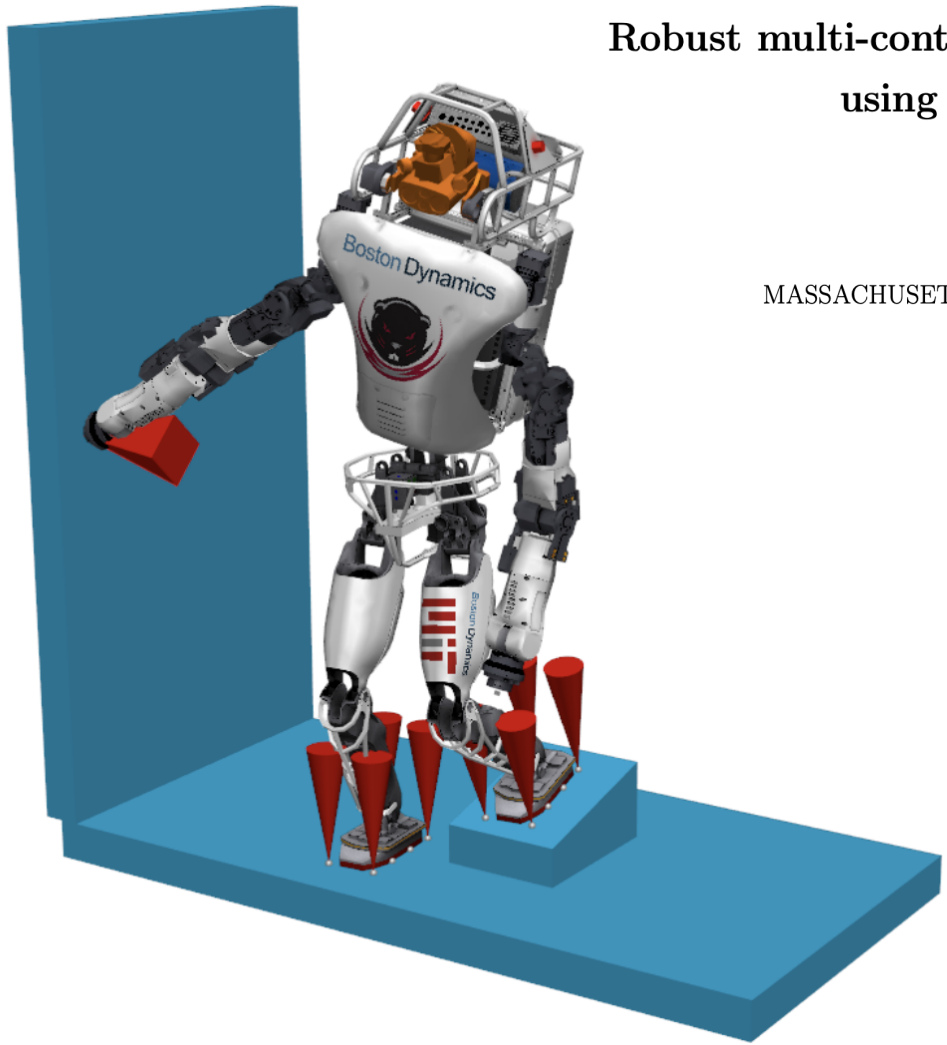


$$A \oplus B = \{a + b \mid a \in A, b \in B\}$$

The red figure is the Minkowski sum of blue and green figures.

https://en.wikipedia.org/wiki/Minkowski_addition

https://manipulation.csail.mit.edu/clutter.html#wrench_cone_algebra



Robust multi-contact dynamical motion planning using contact wrench set

by

Hongkai Dai

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

September 2016

Grasp Pose Detection in Point Clouds

The International Journal of
Robotics Research
2017, Vol. 36(13–14) 1455–1473
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sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0278364917735594
journals.sagepub.com/home/ijr
SAGE

Andreas ten Pas[1], Marcus Gualtieri[1], Kate Saenko[2] and Robert Platt[1]

Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics

Jeffrey Mahler*, Jacky Liang*, Sherdil Niyaz*, Michael Laskey*, Richard Doan*, Xinyu Liu*,
Juan Aparicio Ojea†, and Ken Goldberg*
*Dept. of EECS, University of California, Berkeley

2018 IEEE International Conference on Robotics and Automation (ICRA)
May 21-25, 2018, Brisbane, Australia

Robotic Pick-and-Place of Novel Objects in Clutter with Multi-Affordance Grasping and Cross-Domain Image Matching

Andy Zeng[1], Shuran Song[1], Kuan-Ting Yu[2], Elliott Donlon[2], Francois R. Hogan[2], Maria Bauza[2], Daolin Ma[2],
Orion Taylor[2], Melody Liu[2], Eudald Romo[2], Nima Fazeli[2], Ferran Alet[2], Nikhil Chavan Dafle[2], Rachel Holladay[2],
Isabella Morona[2], Prem Qu Nair[1], Druck Green[2], Ian Taylor[2], Weber Liu[1], Thomas Funkhouser[1], Alberto Rodriguez[2]
[1]Princeton University [2]Massachusetts Institute of Technology

https://www.youtube.com/embed/kfe5bNt35ZI?mute=1&enablejsapi=1&rel=0

https://www.youtube.com/embed/6fG7zwGfIkI?enablejsapi=1&mute=1&start=102&end=160&rel=0

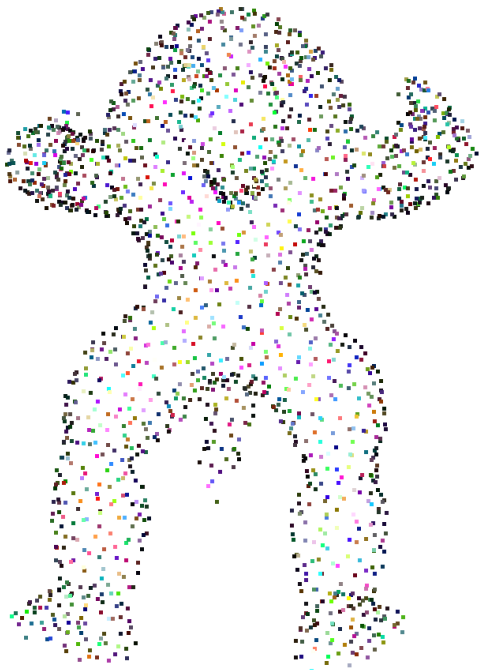


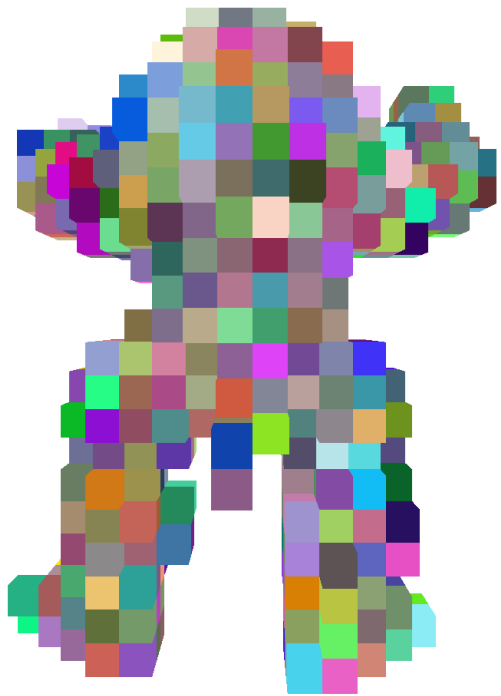
(unpublished) work by Lucas Manuelli ~2017. No deep learning -- this is geometry only.

https://manipulation.csail.mit.edu/data/mustard_bottle_point_clouds.html

https://shainarace.github.io/LinearAlgebra/pca.html

Voxelization



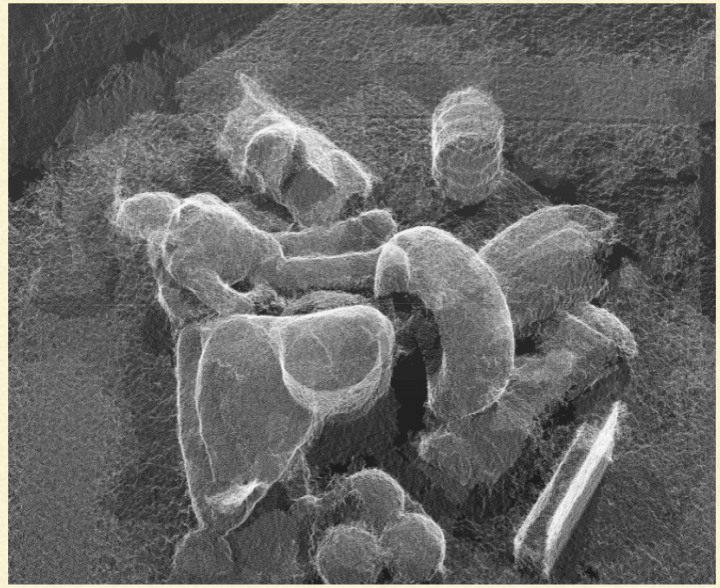




http://www.open3d.org/docs/latest/tutorial/Advanced/voxelization.html

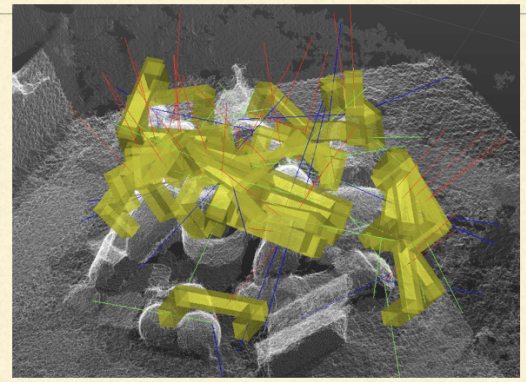
ALGORITHM: SENSOR DATA

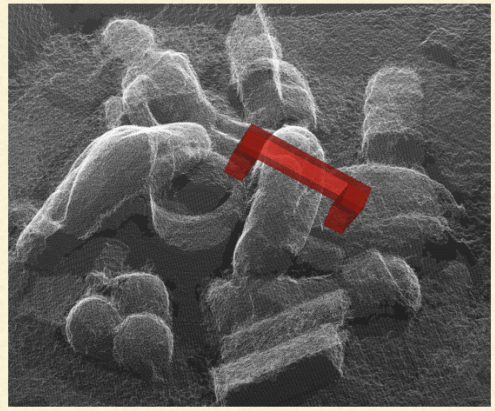
- Algorithm requires **complete pointclouds**.
- Outlier removal
- Normal estimation using camera origins
- Must be **well calibrated** (both intrinsics and extrinsics) in order to provide accurate geometry information to planner



ALGORITHM: SCORING GRASPS

- Construct a scoring function based on antipodal criterion.
- Reward grasps that have contact patch normals aligned with closing direction of gripper.
- Reward grasps that have a larger contact patch.
- In practice this always chooses sensible grasps.

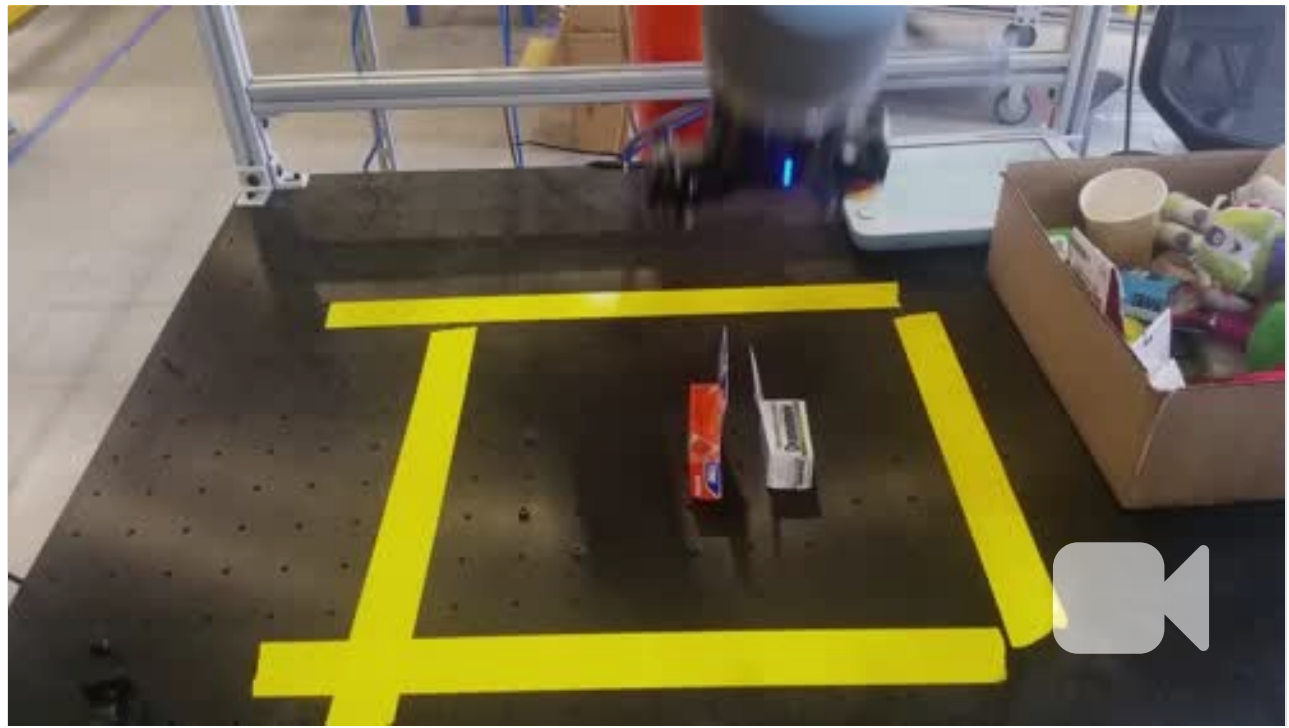




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Limitations of using geometry only

- No understanding of what an object is.
  - "Double picks"
  - Might pick up a heavy object from one corner
- Partial views
- Depth returns don't work for transparent objects
- ...



WHAT GRASPING CAN'T DO

- The presented techniques cannot
  - Identify a specific object
  - Grasp specific object
  - Place object in specific configuration
  - Manipulate object to desired configuration
  - Use a tool
  - Put away the dishes
  - and so much more!





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